

In this chapter we will learn how to use numbers to display different kinds of information

9.1 AVERAGE

9.1.1 Define an average (arithmetic mean)

Consider the following example.

Example. Once Shahid Afridi made 6 runs in the first over, 10 runs in the second over, 8 runs in the third over and 4 runs in the fourth over.

Now answer the following questions:

1. What was the number of his total runs?
 $6 + 10 + 8 + 4 = \mathbf{28 \text{ runs}}$
2. How many overs did he play?
 $\mathbf{4 \text{ overs}}$
3. What was his run rate (runs per over)?
To answer this question we find average as under:

$$\begin{aligned} \text{Run rate} &= \frac{\text{Total runs made in all the overs}}{\text{Total numbers of overs played}} \\ &= \frac{\overset{7}{\cancel{28}} \text{ runs}}{\underset{1}{\cancel{4}} \text{ overs}} = 7 \text{ runs per over} \end{aligned}$$

The run rate **7** is the average score. This 7 represents his performance in all the overs.

Or

Average is the, overall representative value of the information. It is also called the **mean** or the **arithmetic mean**.

Average = Sum of the quantities divided by the total number of quantities

Teacher's Note

Teacher should help the students to understand the concept of average and develop the formula to find the average.

9.1.2 Find an average of given numbers.

Example. Find the average of 5, 8, 10, 12 and 20.

Solution:

Sum of given numbers = $5 + 8 + 10 + 12 + 20$

$$\text{Average} = \frac{\text{Sum of quantities}}{\text{Total number of quantities}}$$

$$\begin{aligned} \text{Average} &= \frac{5 + 8 + 10 + 12 + 20}{5} = \frac{55}{5} \\ &= \frac{\overset{11}{\cancel{55}}}{\underset{1}{\cancel{5}}} = \frac{11}{1} = 11 \end{aligned}$$

Thus the average of the given numbers is **11**.

**Activity 1**

Find the average of first five even numbers.

Solution: First five even numbers are: 2, 4, 6, 8 and 10.

$$\text{Average} = \frac{\text{Sum of numbers}}{\text{Total number}}$$

Solve it,

$$= \frac{2 + 4 + 6 + 8 + 10}{5} = \boxed{\quad} = \boxed{\quad}$$

**Activity 2**

Find the average of first five odd numbers.

Solution: First five odd numbers are: 1, 3, 5, 7 and 9.

$$\text{Average} = \frac{\text{Sum of numbers}}{\text{Total number}}$$

Solve it,

$$= \frac{\quad}{\quad} = \boxed{\quad} = \boxed{\quad}$$

Example: Find the average of given numbers.

4.5, 5.5, 7.5, 6.5, 6.5, 9.5, 7.6

$$\begin{aligned}
 \text{Solution: Average} &= \frac{\text{Sum of given quantities}}{\text{Numbers of given quantities}} \\
 &= \frac{4.5 + 5.5 + 7.5 + 6.5 + 6.5 + 9.5 + 7.6}{7} \\
 &= \frac{47.6}{7} = 6.8
 \end{aligned}$$

EXERCISE 9.1

Find the average (mean) of the following numbers.

- (1) 12, 14, 16, 18 and 20 (2) 1, 2, 3, 4, 5, 6 and 7
 (3) 6, 7, 8, 9, 7, 6, 5 and 15 (4) 2, 3, 5, 7, 11, 13, 17 and 19
 (5) $\frac{1}{2}$, $\frac{2}{5}$, $\frac{3}{4}$ and $\frac{4}{5}$ (6) $\frac{3}{10}$, $\frac{7}{20}$, $\frac{11}{30}$, $\frac{13}{40}$ and $\frac{17}{50}$
 (7) 1.1, 2.2, 3.3, 4.4, 5.5 and 6.6 (8) 13.5, 7.5, 6.5, 9.5, 5.5 and 7.5
 (9) $1\frac{2}{3}$, $4\frac{5}{6}$, $7\frac{8}{9}$, $10\frac{11}{12}$ and $13\frac{14}{15}$ (10) $6\frac{5}{6}$, $7\frac{1}{2}$, $8\frac{1}{3}$ and 5

9.1.3 Solve real life problems involving average.

Example 1. The attendance of class V during six days of a week is 44, 40, 37, 42, 35 and 36. Find the average attendance?

Solution: Daily attendance = 44, 40, 37, 42, 35, 36.

$$\text{Average} = \frac{\text{Total attendance}}{\text{Number of days}}$$

$$\begin{aligned}
 \text{Average attendance} &= \frac{44 + 40 + 37 + 42 + 35 + 36}{6} = \frac{234}{6} \\
 &= \frac{39}{1} = 39
 \end{aligned}$$

Thus average attendance is 39.

Example 2. The business train covers a distance of 450 km in 6 hours. What is its average speed?

Solution:

$$\begin{aligned}\text{Average speed} &= \frac{\text{Distance covered}}{\text{Number of hours}} = \frac{450 \text{ km}}{6 \text{ hours}} \\ &= \frac{\overset{75}{\cancel{450}} \text{ km}}{\underset{1}{\cancel{6}} \text{ hours}} = \frac{75 \text{ km}}{1 \text{ hr}} = 75 \text{ km per hr}\end{aligned}$$

Thus average speed of the train will be **75 km/hr**.

EXERCISE 9.2

1. Rabia obtained 65 marks in Maths, 72 marks in Urdu, 60 marks in Science, 75 marks in Sindhi and 70 marks in Islamiyat. Find her average marks.
2. Kalsoom saved rupees 13, 15, 12, 20, 25, 30 and 18 during a week. What is her average saving per day?
3. The maximum temperature of Sanghar city recorded during the week of June last year was 36.3°C , 42.7°C , 41.6°C , 38.5°C , 40.4°C , 41.9°C and 42.8°C . Find the average temperature during that week.
4. A Qari wants to recite the Holy Qura'n in 15 days during the holy month of Ramzan. What will be the average number of Paras he has to recite daily?
5. Saleem scored 50, 70, 100, 60 runs in the four one day matches. What was his average score?
6. A labour earns Rs 577 on the first day, Rs 600 on the second day and Rs 725 on the third day. Find his average income per day.

7. A train covers a distance of 560 km in 8 hours from Karachi to Rohri. What is its average speed?
8. A car covers a distance of 55 km in the first hour, 60 km in the second hour, 45 km in the third hour and 30 km in the fourth hour. Find the average speed of the car.
9. Akmal gets 8 wickets for 72 runs. Find the average runs he has given for each wicket.
10. Kulsoom's father told her that he would buy her a bicycle if she scored an average of 80 marks in her exams. The scores of all of her exams are below. Did she get the bicycle?

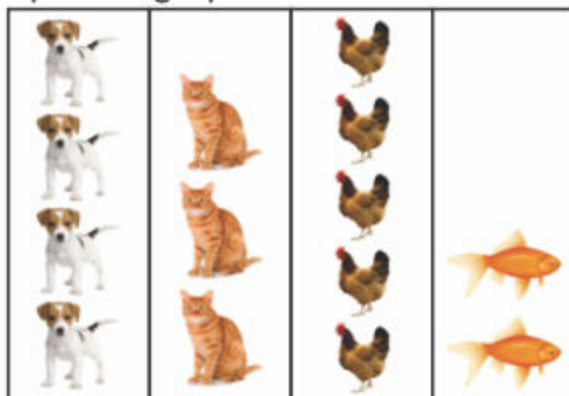
Urdu: 85, English: 70, Math: 82, Science: 77,
Social Studies: 75, Islamiyat: 89.

9.2 BLOCK, COLUMN AND BAR GRAPH

Draw block graphs or column graphs

We have learnt picture graph in previous classes. Let us revise an example.

The number of animals in Ali's farm house has been represented in picture graph.



Dogs Cats Hens Fish

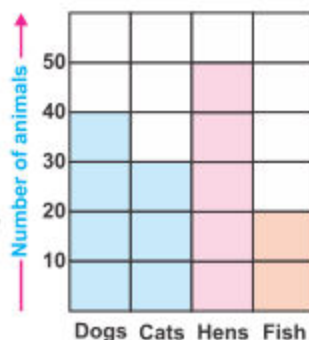
Each  represent 10 dogs.

Each  represent 10 cats.

Each  represent 10 hens.

Each  represent 10 fish.

This diagram is called **picture graph**. Picture graph helps us to see quantity of each thing/item at a glance and help us to compare their differences. We can show these number of animals in blocks as:



We call this a **block graph** or **column graph**. Column graph is used to represent small objects and quantities.

Example 1. Following table shows the result of the favourite breakfast food taken by class V students. Draw a block graph.

Fresh Fruit	Toast/Bread	Eggs	Paratha	Cereal
10	5	15	10	5

Solution:



Read the above graph and answer the following questions:

- How many students eat Paratha?
- Which breakfast is the most favourite?
- Which breakfast is least favourite?

**Activity**

The ages of Nazia's family members are as under. Help her draw the block graph.

Majeed is 30 years old.

Razia is 25 years old.

Nazia is 10 years old.

Hamza is 5 years old.

Step 1: Select the ages on vertical side and family members on horizontal side of the graph.

Step 2: Colour the columns to show the information.

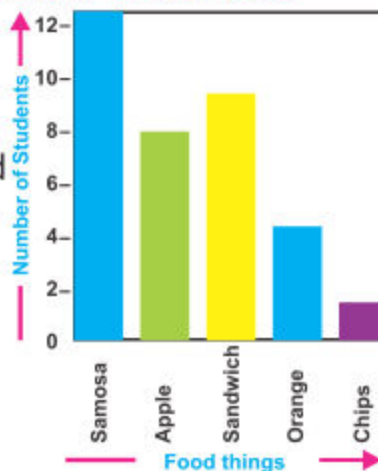


9.2.2 Read and interpret a simple bar graphs given in horizontal and vertical form.

**Activity**

Read the vertical bar graph and write the correct answer in the box.

This bar graph represents information about students and what they ate at break time in school. The different food things are shown horizontally and number of students are shown vertically.



Answer the following questions.

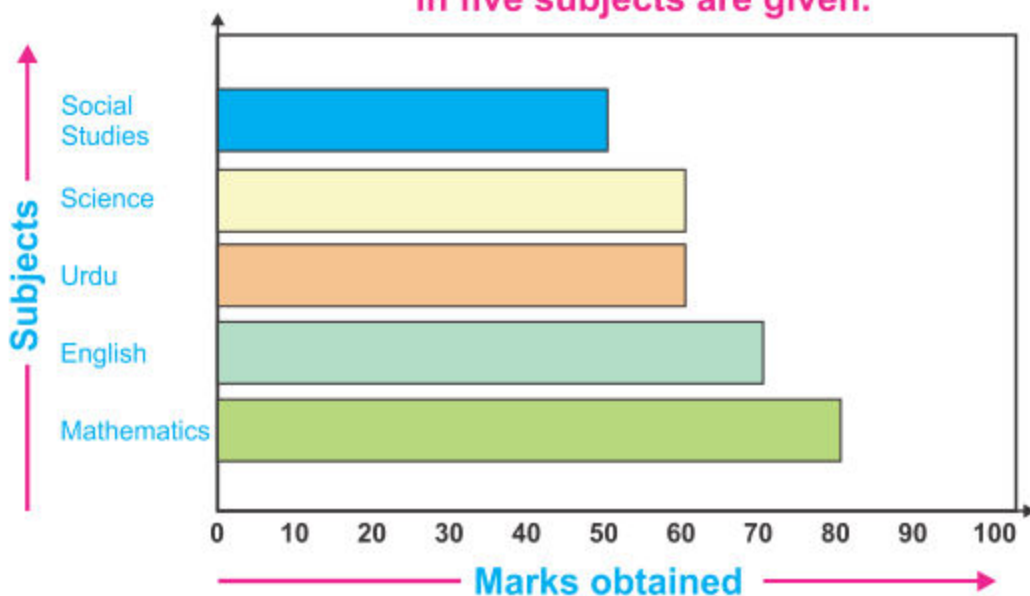
- (1) How many children ate oranges?
- (2) How many children ate apples?
- (3) How many total children were in the school?
- (4) How many children ate samosa?
- (5) How many total ate chips?
- (6) How many children ate sandwich?

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Activity

Read the following horizontal bar graph. In this graph, the marks obtained by Yasir of Grade V in five subjects are given.



Read the horizontal bar graph and answer the following questions:

- How many marks did Yasir get in English?
- How many marks did he get in Urdu?
- How many marks did he get in Mathematics?
- How many marks did he get in Science?
- How many marks did he get in Social studies?
- In which subjects did he got the least number of mark
- How many total marks did he get?
- In which subject did he get the highest marks?
- Which subjects he get equal marks?

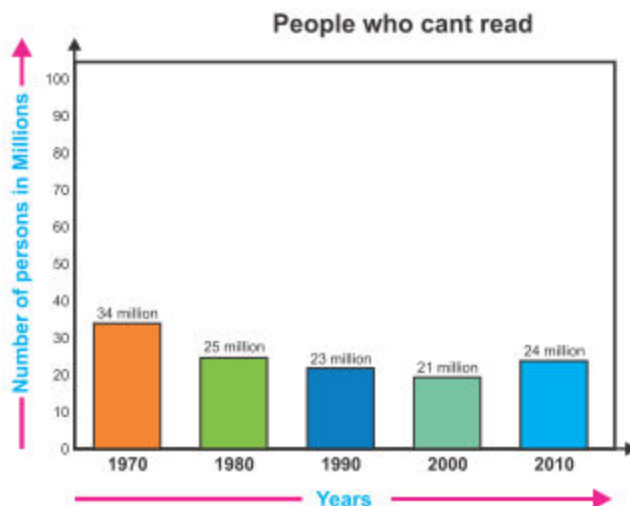
70



Activity

The following bar graph shows the number of people who cannot read of a country in different years.

Time in years is shown on horizontal axis. The number in millions is shown on vertical axis.



Read the vertical bar graph and answer the following questions:

- | | | |
|-------|--|--|
| (i) | How many people could not read in 1990? | <input type="text" value="23 millions"/> |
| (ii) | In which year was number of people who could not read highest? | <input type="text"/> |
| (iii) | What was the number of people who could not read in 1980? | <input type="text"/> |
| (iv) | How many people who could not read were in 2010? | <input type="text"/> |
| (v) | In which year was 2nd highest people who could not read? | <input type="text"/> |
| (vi) | How many fewer people illiterates in 2000 than in 1990? | <input type="text"/> |

EXERCISE 9.3

1. Draw a block graph or column graph of the following:

(i) Attendance of Faraz's class during a week.

Days	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Children	32	35	30	30	25	20

(ii) Result of examination of class V.

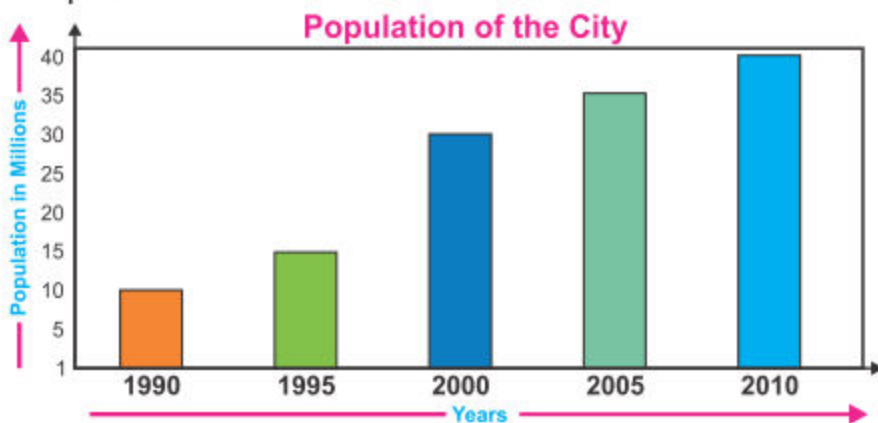
Grade	A1	A	B	C	D
No. of Children	20	25	15	20	5

(iii) The marks obtained by Amjad during annual examination.

Subject	Islamiyat	English	Urdu	Maths	S.Studies	Science
Marks Obtained	80	50	65	90	40	55

2. The population of Dadu city is represented in the following bar graph.

Years are shown on horizontal axis.
Population in millions is shown on vertical axis.

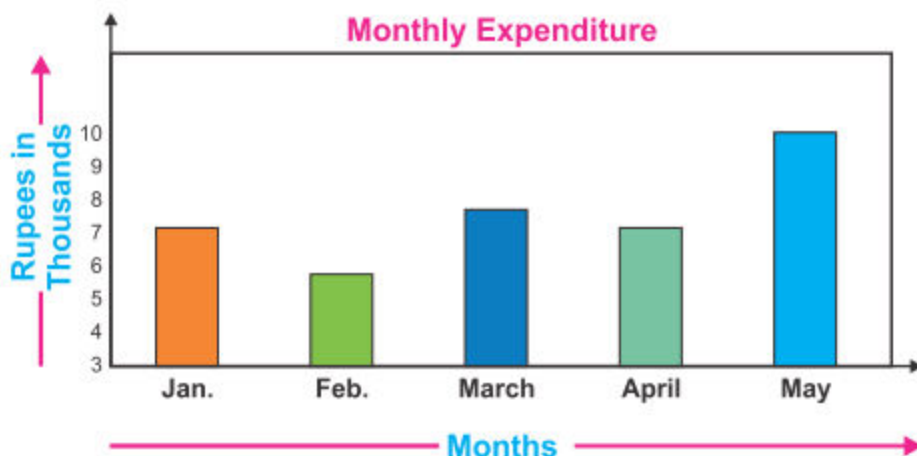


Read the graph and answer the following questions:

- (i) What was the population of the city in 1995?
- (ii) What was the population of the city in 2000?
- (iii) What was the population of the city in 2005?
- (iv) In which year the population was least?
- (v) In which year the population was highest?
- (vi) How many more people were there in 2010 than in 2005?

3. The expenditure of a family for five months is reported by the following bar graph.

Names of months are shown on horizontal axis. Amount of expenditure in thousand rupees is shown on vertical axis.



Look at the graph and answer the following questions:

- (i) In which month expenditure was the least?
- (ii) In which month the expenditure was the greatest?
- (iii) What was the expenditure in the month of February?
- (iv) What was the expenditure in the month of April?
- (v) In which months the expenditure is equal?
- (vi) In which month the expenditure is Rs 8000?
- (vii) What is the highest amount of expenditure?
- (viii) What is the difference of expenditure between months of April and May?

9.2.3 Define and organize data.

(a) Definition of Data

The information collected from any survey, observation or research is called data. Mostly it is represented in the form of numbers.

Example: Favourite colour of class IV students, different age groups in class V, marks obtained by some one in an examination, any office record, match scores; etc.

(b) Organizing Data

There are many ways of collecting and organizing data.

Data can be represented through pictures, tables, charts and graphs.

Example: The organizer of a market survey wants to know the ages of the customers.

Solution:

The ages of customers are:

17, 12, 15, 20, 25, 17, 19, 11, 12,
15, 20, 20, 12, 20, 22, 20, 19, 15

This way of representing the information does not help the organizer to answer any question.

Now we write the above information in ascending order:

11, 12, 12, 12, 15, 15, 15, 17, 17,
19, 19, 20, 20, 20, 20, 20, 22, 25

Table (i)

Teacher's Note

Teacher should help the students to develop the concept of data and of organizing different data. He/she should give them tasks to collect and represent different data from their surroundings.

Though it is one way but it is time taking. There is possible that some data may be missed. However, it is easier to read data.

Now we organize the data in another way through tally method.

Age of customers in years	Tally marks	Number of customers
11		1
12		3
15		3
17		2
19		2
20		5
22		1
25		1

Table (ii)

The organizer now read the ages one by one from Table (i) and draws one small line called tally in the tally column against the age read in table (ii). If a number occurs five times he crosses the four tallies already drawn by a slash like this (||||). Thus (||||) stands for five numbers. Then counts the tallies and write the numbers in the last column shown in table.

Example:

The mathematics marks of class V students are as under:

organize the data

32, 30, 31, 33, 33, 34, 34, 33, 32, 32, 34, 35,
35, 32, 33, 32, 31, 33, 32

Solution:

Now we write the above information in ascending order:

30, 31, 31, 32, 32, 32, 32, 32, 32, 33,
33, 33, 33, 33, 34, 34, 34, 35, 35

The data is represented through tally chart.

Marks obtained	Tally	No. of students
30		1
31		2
32		6
33		5
34		3
35		2



Activity

Organize the following information through tally method. The ages of students of class V in a school are given in the following table:

Ages of students in years	11	12	11	13	10	11	13	12	13	14
	12	13	11	11	14	10	12	12	11	13

Solution: Writing the above information in ascending order:

Ages of students in years	10		11						12	
				13					14	

Organize the given data through tally method

Ages of students in years	Tally	No. of same ages of students
10		
11		
14		

EXERCISE 9.4

Organize the following survey information through tally method.

1. The pocket money of students of Class V is found as:
Rs 30, Rs 40, Rs 50, Rs 40, Rs 50, Rs 60, Rs 50, Rs 40,
Rs 70, Rs 60, Rs 50, Rs 40, Rs 30, Rs 40 and Rs 50.
2. The heights of students of Class V in a school are noted
as: 110 cm, 115 cm, 100 cm, 105 cm, 104 cm, 110 cm,
108 cm, 100 cm, 100 cm, 105 cm, 110 cm, 104 cm,
105 cm, 100 cm, 115 cm, 105 cm, 110 cm, 105 cm,
100 cm and 105 cm.
3. The attendance of 20 days of class V students are: 15,
18, 16, 17, 18, 17, 18, 17, 16, 15, 16, 17, 18, 18, 17, 16,
18, 17, 16 and 17.
4. A gardener planted different numbers of plants in 18 days
as under:
2, 1, 3, 5, 6, 3, 2, 3, 1, 2, 3, 4, 3, 2, 5, 5, 2 and 3.
5. The number of patients daily checked by doctor in 15
days are: 35, 40, 36, 35, 35, 40, 37, 35, 40, 38, 35, 40,
39, 40 and 35.
6. A team played 6 matches. The score was 210, 189, 180,
205, 175 and 165 runs respectively. What is the average
score?
7. The enrolment of a school is 460 students. If there are
8 class rooms, what is the average of students in each
class room?

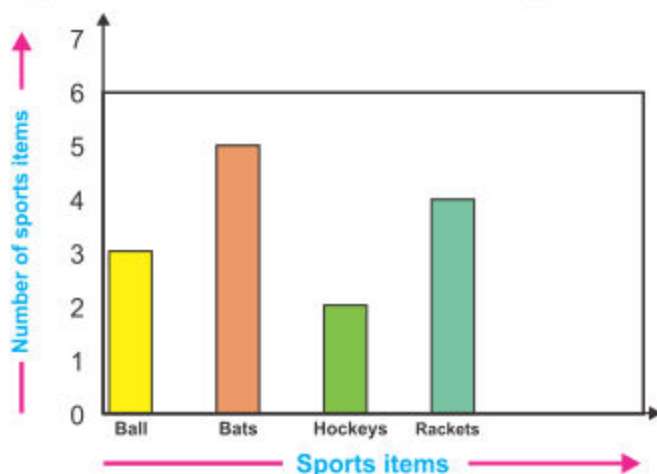
REVIEW EXERCISE 9

- Find the average of following numbers:
 - 4, 3, 5, 8, 6, 10.
 - 10, 20, 30, 40, 50
- The six hours journey of a person on car is given below.

Hours	1	2	3	4	5	6
Distance in kilometres	20	18	20	22	16	18

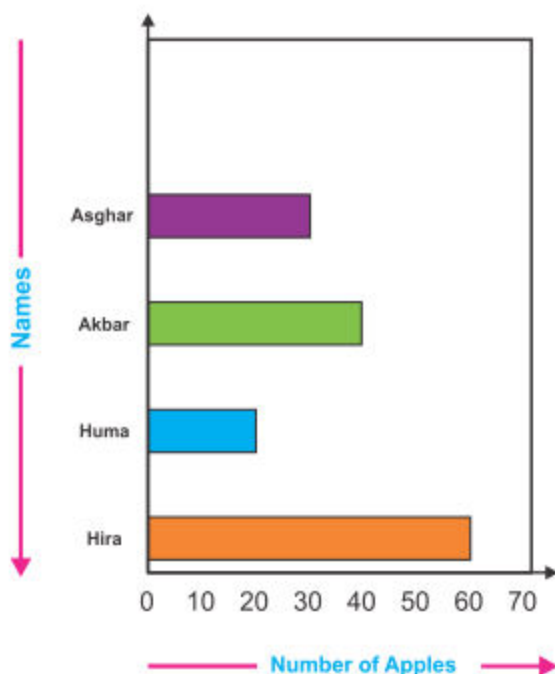
What is the average distance covered per hour?

- A batsman made **160** runs in two innings. Find the average runs he made in each innings.
- Read the graph and answer the following questions.



- How many balls are there?
- How many bats are there?
- How many hockeys and bats are there?
- How many more rackets are there than hockeys?
- How many fewer balls are there than bats?
- How many bats and rackets are there ?

5. Interpret the following vertical bar graph.



- (i) How many apples does Asghar have?
- (ii) How many apples does Akbar have?
- (iii) How many apples does Huma have?
- (iv) How many apples does Hira have?
- (v) How many apples do Asghar and Akbar have?
- (vi) How many apples do Huma and Hira have?

6. Fill in the blanks.

- (i) The average of 5, 15, 30, 10 and 20 is _____
- (ii) The average of 20, 40, 30 and 35 is _____
- (iii) When we choose a suitable symbol to represent each part of information, we will use _____ graph.

GLOSSARY

Acute angle:	An angle whose measure lies between 0° and 90° .
Acute angled triangle:	A triangle in which all angles are acute.
Adjacent angles:	Two angles with a common vertex and a common arm are called adjacent angles.
Angle:	The amount of turning between two arms about a common point.
Arc:	A part of the circumference of a circle.
Area:	The space occupied within the boundary of a shape is called an area.
Associative property of addition:	The property that when any three numbers (fractions) are added in any order, their sum is always the same.
Associative property of multiplication:	The property that when any three numbers (fractions) are multiplied in any order, their sum is always the same.
Average:	Sum of quantities divided by the total number of quantities.
Bar graph:	It represents each part of the information in the form of bars (vertical or horizontal).
Block graph:	The graph in which we choose a suitable symbol to represent each part of the information.
Capacity:	The amount of liquid a container can hold.
Centimetre:	A unit of length, $100 \text{ centimetres (cm)} = 1 \text{ metre (m)}$
Circle:	The set of all points which are equidistant from the fixed point, is called a circle
Commutative property of Multiplication:	The property in which any two numbers fractions when multiplied to each other in any order, their product is always same
Commutative property of addition:	The property that when any two numbers (fractions) are added in any order their sum is always same.
Complementary angles:	Two angles whose sum of the measures is equal to 90° .
Data:	Information presented in the form of numbers.
Decimal:	Any number containing a fractional part indicated by a decimal point is called decimal number or decimal.
Decimal fraction:	A fraction which is expressed in decimal point.

GLOSSARY

Denominator:	Lower part of the common fraction.
Diametre:	A line segment that divides the circle into two equal halves. 
Direct proportion:	The relationship between two ratios in which increase in one quantity causes a proportional increase in the other quantity and decrease in one quantity causes a proportional decrease in the other quantity.
Dividend:	A number which is to be divided by another number.
Divisibility:	A division in which when a number is divided by another, the remainder is zero.
Divisor:	A number which can divide the other number exactly.
Edge:	A one dimensional line segment joining two vertices.
Equivalent fractions:	Fractions that have the same lowest form.
Equilateral triangle:	A triangle in which all the three sides are equal in length.
Factors:	The divisor of a number.
Factorization:	A number represented as a product of its factors.
Fractions:	Part of a whole.
Gram:	Unit of mass.
Graph:	A pictorial representation of a data.
GCD:	Greatest Common Divisor.
Hours:	24 th part of the day, 60 minutes. A unit of time 1 hour = 60 minutes
HCF:	Highest Common Factor.
Inverse proportion:	The relationship between the two ratios in which increase in one quantity causes a proportional decrease in the other quantity and a decrease in one quantity causes a proportional increase in the other quantity.
Isosceles triangle:	A triangle with its two sides equal in length.
Kilogram:	A unit of mass. 1 kilogram (kg) = 1000 grams (g)
L.C.M	Least Common Multiple.
Like decimals:	The decimals having same number of decimal places.

GLOSSARY

Like fractions:	Fraction having same denominator.
Line:	 This figure represents a line AB.
Line segment:	Shortest distance between two points. 
Litre:	Unit of volume/capacity 1 litre (ℓ) = 1000 millilitres ($m\ell$)
Lunar Calendar:	(Hijrah Qamri Calendar) Islamic Calendar in a solar year.
Mass:	Quantity of matter present in a body.
Millilitre:	Thousandth part of a litre.
Millimetre:	Thousandths part of a metre.
Million:	The smallest seven digit number i.e. 1,000,000 (Ten hundred thousand).
Minute:	Sixtieth part of an hour. 1 minute = 60 seconds
Mixed fraction:	A fraction contains both a whole number and a proper common fraction.
Month:	A unit of time. 1 month = 30 days
Numerator:	Upper number of common fraction.
Obtuse angle:	An angle whose measure lies between 90° and 180° .
Obtuse angled triangle:	A triangle which has one of its angles obtuse angle.
Percentage:	The word percent is a short form of the Latin word "Percentum". Percent means out of hundred or per hundred.
Perimeter:	The distance along the sides or boundary of a closed shape.
Prime factorization:	A factorization in which every factor is a prime factor.
Proper fraction:	A fraction whose numerator is less than the denominator.
Proportion:	The quality of two ratios.
Quadrilateral:	A four sided closed figure.
Quotient:	The number shows how many times the divisor has been repeatedly subtracted from dividend.
Ratio:	A comparison of two quantities of the same kind.
Radius:	The distance from the centre of the circle to the boundary of the circle.

GLOSSARY

Ray:	An arrow mark on one end point of a line segment 
Rectangle:	A quadrilateral whose opposite sides are equal and have four right angles.
Reflex angle:	An angle of measure greater than 180° but less than 360°
Remainder:	The number left over when one integer is divided by another.
Right angle:	An angle whose measure is 90° .
Right angled triangle:	A triangle which has one of its angle of the measure 90° .
Round off decimals:	To round off a decimal nearest to the whole number, check the first decimal place and accordingly round off the number.
Scalene triangle:	A triangle whose all sides are of different measures.
Second:	Unit of time, $\frac{1}{60}$ the part of a minute.
Solar Calendar:	In this calendar, the dates indicates the position around the sun (365 days in a year).
Square:	A quadrilateral whose all four sides are equal and has four right angles.
Straight angle:	An angle whose measure equals to 180° .
Subtraction:	Symbol (-). The process of finding the difference between two numbers/quantities.
Supplementary angles:	Two angles whose sum of the measures is equal to 180° .
Symbol:	A sign or notation used to represent an operation, element or relation etc
Temperature:	A quantity that measure the hotness or coldness of a body or substance.
Triangle:	A three sided closed geometrical figure.
Unit fraction:	Numerator is equal to the denominator.
Unitary method:	The process of finding the price of one (unit) item, from which we find the price of a number of similar items.
Unlike decimals:	The decimals having different number of decimal places.
Unlike fractions:	Fractions whose denominators are not same.
Vertex:	An angular point of any shape.

ANSWERS

EXERCISE 1.1

H-M	T-M	M	H-th	T-th	Th	H	T	O
1	7	1	2	8	3	6	8	0

- A.** So, the number in words is
One hundred seventy one million, Two hundred eighty three thousands and six hundred eighty
- B.** (1) 45,672 (2) 2,670,273 (3) 34,296,127 (4) 100,000,000
- C.** (1) Sixty six millions, Six hundred fifty five thousands and five hundred twenty two.
(2) Ninety six millions, three hundred forty thousands and five hundred twenty nine.
(3) Two hundred forty five millions, six hundred seventy two thousands and three hundred sixteen.
(4) One hundred millions.
- D.** (1) 1,002,600 (2) 9,099,077 (3) 58,862,045
(4) 1,000,000,000 (5) 345,671,806

EXERCISE 1.2

- A.** (1) 981,802 (2) 4,632,048 (3) 8,623,037 (4) 7,051,459
(5) 4,531,777 (6) 6,097,173 (7) 8,928,449 (8) 3,640,193
(9) 9,500,950 (10) 37,314,112 (11) 79,613,418 (12) 789,451,507
- B.** (1) 1,243,731 (2) 3,773,535 (3) 40,811,696 (4) 88,856,946
(5) 905,347,265 (6) 921,490,114
- D.** 475,200

EXERCISE 1.3

- A.** (1) 2698782 (2) 5240309 (3) 3200104 (4) 2046948
(5) 49918102 (6) 57095640
- B.** (1) 386121 (2) 2911201 (3) 920927 (4) 38151829
(5) 53905409 (6) 521889785
- D.** 46,935

ANSWERS

EXERCISE 1.4

- A. (1) 41360 (2) 345690 (3) 210340 (4) 1534700
(5) 2779600 (6) 15543000 (7) 41357000 (8) 386975000
- B. (1) 97100 (2) 5086240 (3) 1792560 (4) 29428650
(5) 2562000 (6) 168549000
- C. (1) 852687 (2) 20934034 (3) 4890300 (4) 43038336
(5) 1680602 (6) 13135892 (7) 18538128 (8) 36112216
(9) 5471025 (10) 609922660 (11) 173941344 (12) 463038004
(13) 140699670 (14) 522851472 (15) 280170566 (16) 384251808
- D. 14,625

EXERCISE 1.5

- A. (1) 8,965 (2) 5,568 (3) 6,589 (4) 9,608
(5) 562 (6) 659 (7) 1,377 (8) 568
- B. (1) remainder = 8, quotient = 567 (2) remainder = 5, quotient = 39678
(3) remainder = 5, quotient = 4734 (4) remainder = 16, quotient = 8432
(5) remainder = 6, quotient = 52301 (6) remainder = 879, quotient = 8256
(7) remainder = 782, quotient = 6456 (8) remainder = 0, quotient = 9650
- C. 123

EXERCISE 1.6

- A. (1) 199624 rupees (2) Rs 2700 (3) Rs 114 by each student
(4) 7545792 marbles (5) Rs 134502500 (6) 3354150 litres
(7) 11815 rolls required (8) Rs 2434372

EXERCISE 1.7

- A. Step 1: Bracket first Subtract 213 - 123 Step 4: Add 90 + 300
Step 2: Divide 64 ÷ 8 Step 5: Subtract 390 - 8
Step 3: Multiply 60 x 5
- B. (1) 18 (2) 3 (3) 20 (4) 666
(5) 288 (6) 27 (7) 6 (8) 114
(9) 288 (10) 27 (11) 10 (12) 1

EXERCISE 1.8

- B. (1) 5, 3 (2) 32, 32 (3) 30, 40 (4) +, x
(5) 5, + (6) 5, 13 (7) 30, 20, 10 (8) 12, 8, 4, 4

ANSWERS

REVIEW EXERCISE 1

1. (i) Two hundred forty six millions, four hundred sixteen thousands and two hundred seventy nine.
(ii) Nine hundred five millions, four hundred seven thousands and six hundred eight.
2. (i) 75,026,420 (ii) 405,745,806
3. (i) 211,029,597 (ii) 24,137,404 (iii) 737,718,214
4. (i) 211,678,206 (ii) 616,236,553 (iii) 407,288,461
5. (i) 12,430 (ii) 43,305,525 (iii) 2,269,160
(iv) 589,107,636 (v) 405,617,000
6. (i) remainder = 10, quotient = 25113 (ii) remainder = 325, quotient = 757
7. (i) 450 (ii) 26 9. 858935 litres 10. 800 boxes

EXERCISE 2.1

- | | | | | |
|----|--------|---------|---------|--------|
| A. | (1) 4 | (2) 9 | (3) 8 | (4) 14 |
| | (5) 22 | (6) 26 | (7) 20 | (8) 16 |
| | (9) 7 | (10) 13 | (11) 15 | (12) 7 |
| B. | (1) 14 | (2) 22 | (3) 19 | (4) 17 |
| | (5) 16 | (6) 18 | (7) 20 | (8) 12 |
| | (9) 16 | (10) 15 | (11) 14 | (12) 9 |

EXERCISE 2.2

- | | | | | |
|----|----------|----------|----------|----------|
| A. | (1) 108 | (2) 165 | (3) 156 | (4) 240 |
| | (5) 432 | (6) 3600 | (7) 1176 | (8) 396 |
| | (9) 150 | (10) 450 | (11) 160 | (12) 432 |
| B. | (1) 48 | (2) 100 | (3) 144 | (4) 540 |
| | (5) 1260 | (6) 192 | (7) 1080 | (8) 3960 |
| | (9) 240 | (10) 96 | (11) 540 | (12) 240 |

EXERCISE 2.3

- | | | | |
|--------------|---|-------------|----------------|
| (1) 600 | (2) 6 | (3) 30 days | (4) 105 |
| (5) 60 hours | (6) 36 cm | (7) 16 cm | (8) 600 litres |
| (9) 180 cm | (10) Packets of 6 pencils and 5 erasers | | |

REVIEW EXERCISE 2

- | | | | | |
|-------------|---------------|--------------|--------|-------|
| 1. (i) a | (ii) d | (iii) b | (iv) c | (v) c |
| 2. 24 | 3. 6 | 4. 60 apples | | |
| 5. 5 metres | 6. 600 litres | | | |

ANSWERS

EXERCISE 3.1

- A. (1) $\frac{5}{6}$ (2) $\frac{7}{8}$ (3) $\frac{11}{15}$ (4) $\frac{17}{24}$ (5) $\frac{35}{16}$ (6) $\frac{29}{35}$
- B. (1) $\frac{29}{45}$ (2) $\frac{13}{45}$ (3) $\frac{13}{48}$ (4) $2\frac{2}{5}$ (5) $4\frac{9}{20}$ (6) $2\frac{31}{96}$
- C. (1) $\frac{5}{12}$ (2) $\frac{1}{4}$ (3) $\frac{1}{3}$ (4) $\frac{1}{3}$ (5) 3 (6) $\frac{7}{10}$
- D. (1) $\frac{7}{24}$ (2) $\frac{1}{4}$ (3) $\frac{2}{7}$ (4) $\frac{1}{4}$ (5) $-\frac{1}{2}$ (6) $-2\frac{4}{5}$
 (7) $-2\frac{11}{18}$ (8) $-2\frac{1}{3}$ (9) $\frac{1}{2}$
- E. (1) $\frac{7}{10}$ (2) $\frac{1}{2}$ (3) $\frac{11}{42}$ F. (1) $\frac{1}{12}$ (2) $\frac{7}{15}$ (3) $\frac{1}{20}$

EXERCISE 3.2

- A. (1) 

So, $\frac{3}{4} \times 4 = 3$

- (2) 1 (3) 3 (4) 1 (5) 2 (6) 2 (7) 2 (8) 4 (9) 6

Note: Compare the answer with drawn figures.

- B. (1) $\frac{1}{10}$ (2) $\frac{3}{20}$ (3) $\frac{2}{9}$ (4) $\frac{15}{16}$ (5) $1\frac{13}{15}$ (6) $5\frac{1}{4}$
- C. (1) $\frac{2}{3}$ (2) $2\frac{1}{2}$ (3) $1\frac{1}{2}$ (4) $3\frac{1}{15}$ (5) $5\frac{5}{12}$ (6) $11\frac{1}{9}$

EXERCISE 3.3

- (1) 6 (2) 2 (3) 3 (4) 4 (5) 4 (6) 5
 (7) 25 (8) 14 (9) 4 (10) 10 (11) $12\frac{1}{2}$ (12) $2\frac{1}{2}$

EXERCISE 3.5

- (1) $\frac{1}{3}$ metres (2) 495 students (3) 8 drips (4) 20 kg
 (5) Rs 1482 (6) $1\frac{4}{5}$ metres (7) 20 litres (8) 18 metres
 (9) Rs 705 (10) 110 metres

ANSWERS

EXERCISE 3.6

- A. (1) $\frac{1}{2}$ (2) $\frac{1}{4}$ (3) 3 (4) 5 (5) 10
(6) $\frac{4}{3}$ (7) $\frac{7}{20}$ (8) $\frac{5}{13}$ (9) $\frac{3}{7}$ (10) $\frac{3}{14}$
- B. (1) $\frac{3}{5}$ (2) $\frac{1}{6}$ (3) $\frac{1}{8}$ (4) $\frac{1}{10}$ (5) $\frac{1}{12}$ (6) $\frac{1}{15}$
(7) $\frac{1}{16}$ (8) $\frac{1}{27}$ (9) $\frac{1}{15}$ (10) $\frac{1}{18}$ (11) $\frac{1}{25}$ (12) $\frac{2}{3}$

EXERCISE 3.7

- (1) $\frac{2}{3}$ (2) $2\frac{5}{8}$ (3) $\frac{4}{5}$ (4) $\frac{2}{9}$ (5) 2
(6) 1 (7) $1\frac{1}{3}$ (8) $\frac{1}{2}$ (9) $\frac{5}{9}$ (10) $\frac{44}{57}$
(11) $2\frac{3}{7}$ (12) $1\frac{31}{35}$ (13) $4\frac{4}{15}$ (14) $3\frac{1}{2}$ (15) $2\frac{22}{171}$

EXERCISE 3.8

- (1) Rs 20 (2) 7 cars (3) 28 pieces (4) 7 servings
(5) $25\frac{1}{4}$ metres (6) $5\frac{1}{2}$ metres (7) 33 bottles (8) $3\frac{1}{2}$ km

EXERCISE 3.9

- (1) $1\frac{5}{108}$ (2) $\frac{31}{42}$ (3) $1\frac{5}{66}$ (4) $3\frac{5}{6}$ (5) $\frac{5}{8}$ (6) $1\frac{13}{56}$
(7) $3\frac{17}{27}$ (8) $-2\frac{7}{27}$ (9) $-2\frac{31}{105}$ (10) $22\frac{61}{72}$ (11) $64\frac{2}{3}$ (12) $17\frac{5}{6}$

REVIEW EXERCISE 3

1. (i) F (ii) T (iii) F (iv) T
(v) T (vi) T (vii) T (viii) T
2. (i) Rs $10\frac{3}{20}$ (ii) Afzal got $\frac{1}{10}$ more (iii) Rs 300
(iv) $\frac{3}{8}$ m left (v) 202 packets (vi) 800 tickets
3. (a) $-\frac{1643}{9000}$ (b) $\frac{15}{16}$

ANSWERS

EXERCISE 4.1

- A.** (1) 28.78 (2) 40.967 (3) 99.67 (4) 257.253
(5) 47.443 (6) 97.475 (7) 1131.84 (8) 711.047
(9) 100.44 (10) 295.357 (11) 322.826 (12) 911.565
- B.** (1) 19.22 (2) 41.165 (3) 51.13 (4) 256.753
(5) 42.103 (6) 605.78 (7) 441.08 (8) 638.087
(9) 31.1 (10) 113.044 (11) 611.11 (12) 130.325

EXERCISE 4.2

- A.** Decimals having (i) One decimal place= 0.6 and 62.1
(ii) Two decimal places= 0.08, 34.25, 3.36 and 38.66
(iii) Three decimal places= 2.123 and 52.302
- B.** (1) 1.75 (2) 17.5 (3) 175 (4) 350.58
(5) 3505.8 (6) 35058 (7) 81.5 (8) 815
(9) 8150 (10) 3244.23 (11) 32442.3 (12) 324423
(13) 0.067 (14) 0.67 (15) 6.7

EXERCISE 4.3

- (1) 0.6675 (2) 0.06675 (3) 0.006676 (4) 3.589
(5) 0.3589 (6) 0.03589 (7) 81.54 (8) 8.154
(9) 0.8154 (10) 0.0085 (11) 0.00085 (12) 0.000085

EXERCISE 4.4

- B.** (1) 3.25 (2) 32.5 (3) 3.9 (4) 21
(5) 2.8 (6) 29.75 (7) 34 (8) 38.25
(9) 1.91 (10) 3.152 (11) 172.06 (12) 48321
(13) 114.75 (14) 9.54 (15) 1211.475 (16) 688.5
(17) 16469.448 (18) 33257.388

EXERCISE 4.5

- A.** (1) 0.13 (2) 1.3 (3) 0.013 (4) 0.6
(5) 0.08 (6) 0.61 (7) 1.2 (8) 4.9
(9) 0.077 (10) 1.399 (11) 5.693 (12) 9.67
- B.** (1) 0.894 (2) 14.94 (3) 32.86 (4) 2.425
(5) 0.013 (6) 0.111 (7) 1.367 (8) 3.052
(9) 174.096

ANSWERS

EXERCISE 4.6

- | | | | |
|-------------|------------|-------------|------------|
| (1) 0.26 | (2) 0.774 | (3) 12.75 | (4) 23.98 |
| (5) 0.15 | (6) 0.006 | (7) 0.44 | (8) 0.21 |
| (9) 0.078 | (10) 0.003 | (11) 0.0001 | (12) 0.088 |
| (13) 0.2322 | (14) 2.125 | (15) 4.1965 | |

EXERCISE 4.7

- A.**
- | | | | | |
|-------------|--------------|--------------|-------------|------------|
| (1) 56.04 | (2) 0.9048 | (3) 4.284 | (4) 81.84 | (5) 0.4824 |
| (6) 17.8068 | (7) 1279.356 | (8) 548.1025 | (9) 925.934 | |
- B.**
- | | | | |
|-------------|--------------|-------------|-------------|
| (1) 0.112 | (2) 0.225 | (3) 0.0046 | (4) 0.014 |
| (5) 0.01625 | (6) 0.0006 | (7) 0.0016 | (8) 0.945 |
| (9) 0.01612 | (10) 0.01254 | (11) 0.0125 | (12) 0.1057 |
- C.**
- | | | | |
|-----------|------------|-------------|------------|
| (1) 3.5 | (2) 2.73 | (3) 2.66 | (4) 0.603 |
| (5) 8.01 | (6) 13.26 | (7) 44.304 | (8) 23.1 |
| (9) 18.33 | (10) 49.83 | (11) 213.18 | (12) 72.22 |

EXERCISE 4.8

- A.**
- | | | | |
|-----------|------------|-----------|---------|
| (1) 3.6 | (2) 24 | (3) 14.7 | (4) 112 |
| (5) 0.547 | (6) 1.09 | (7) 56 | (8) 112 |
| (9) 128 | (10) 27.16 | (11) 2502 | (12) 57 |
- B.**
- | | | | |
|-----------|------------|------------|-----------|
| (1) 0.08 | (2) 0.6 | (3) 0.6 | (4) 0.09 |
| (5) 0.11 | (6) 0.013 | (7) 2.4 | (8) 10.02 |
| (9) 0.03 | (10) 0.017 | (11) 0.012 | (12) 0.6 |
| (13) 25.6 | (14) 1.3 | (15) 0.014 | |

EXERCISE 4.9

- | | | | |
|------------|------------|------------|------------|
| (1) 1.25 | (2) 1.666 | (3) 0.8 | (4) 0.7 |
| (5) 2.142 | (6) 1.555 | (7) 0.625 | (8) 2.555 |
| (9) 4.571 | (10) 3.461 | (11) 3.333 | (12) 0.642 |
| (13) 0.687 | (14) 0.416 | (15) 0.85 | (16) 2.083 |
| (17) 9.423 | (18) 31.25 | (19) 1.2 | (20) 0.92 |

EXERCISE 4.10

- | | | | | |
|-----------|--------------|------------|------------|--------------|
| (1) 5.9 | (2) 123.7175 | (3) 34.06 | (4) 16.63 | (5) 6.015 |
| (6) 79.38 | (7) 46.196 | (8) 26.382 | (9) 21.454 | (10) 194.711 |

ANSWERS

EXERCISE 4.11

- A.** (1) 2 (2) 6 (3) 8 (4) 7 (5) 10 (6) 8
 (7) 8 (8) 50 (9) 59 (10) 78 (11) 82 (12) 100
- B.** (1) 32.4 (2) 25.2 (3) 6.2 (4) 6.4
 (5) 76.8 (6) 95.2 (7) 12.9 (8) 6.0
 (9) 3.4 (10) 11.8 (11) 50.5 (12) 60.2
- C.** (1) 32.39 (2) 25.06 (3) 6.78 (4) 6.42
 (5) 76.80 (6) 8.48 (7) 0.96 (8) 58.19
 (9) 4.01 (10) 40.98 (11) 70.49 (12) 19.02

EXERCISE 4.12

- A.** (1) 0.875 (2) 0.9 (3) 0.4 (4) 0.68
 (5) 1.22 (6) 0.95 (7) 1.225 (8) 0.8875
 (9) 0.902 (10) 0.316 (11) 0.83 (12) 0.2775
 (13) 0.97125 (14) 0.551 (15) 0.0999
- B.** (1) $\frac{1}{2}$ (2) $1\frac{1}{20}$ (3) $3\frac{14}{25}$ (4) $\frac{113}{200}$ (5) $\frac{23}{1000}$ (6) $\frac{1}{4}$
 (7) $\frac{69}{200}$ (8) $35\frac{253}{500}$ (9) $\frac{8}{125}$ (10) $\frac{189}{200}$ (11) $41\frac{5}{8}$ (12) $46\frac{64}{625}$

EXERCISE 4.13

- (1) 3.135 kg weight (2) 102.25 kg more (3) 11.44 g total
 (4) Rs 31993.25 (5) Rs 70372.50 (6) 89.58 m wire
 (7) 12.50 m left (8) 7 dresses (9) 0.12 kg per day
 (10) Rs 156.50

EXERCISE 4.14

- A.** (1) 0.25 (2) 0.3 (3) 0.35 (4) 0.4
 (5) 0.65 (6) 0.7 (7) 0.8 (8) 0.85
 (9) 0.95 (10) 0.99 (11) 1.05 (12) 1.15
- B.** (1) $\frac{4}{5}$, 80% (2) $\frac{17}{50}$, 34% (3) $\frac{14}{25}$, 56% (4) $\frac{63}{100}$, 63%
 (5) $\frac{11}{20}$, 55% (6) $\frac{33}{50}$, 66% (7) $3\frac{9}{20}$, 345% (8) $3\frac{3}{5}$, 360%
 (9) $5\frac{1}{2}$, 550% (10) $2\frac{1}{20}$, 205% (11) $25\frac{1}{2}$, 2550% (12) $55\frac{1}{4}$, 5525%

ANSWERS

- C. (1) 80% (2) 24% (3) 55% (4) 62.5%
(5) 42.5% (6) 50% (7) 30% (8) 60%
(9) 142% (10) 180%

EXERCISE 4.15

- (1) 216 boys (2) 495 students (3) 630 employees (4) 1586 houses
(5) 240 students (6) 90 cars (7) Rs 1501.50

REVIEW EXERCISE 4

- A. (1) 100 (2) 100 (3) 0.65 (4) 4.444 (5) 4.95
B. (1) 0.016, 0.463, 1.995, 2.087 (2) 5.661 (3) 10.09 m higher (5) 0.6
C. (1) 4 dresses (2) 1680 marks (3) (i) 15.75 (ii) 3.42 (iii) 242 (iv) 11

EXERCISE 5.1

- A. (1) 1 km 600 m (2) 2 km 483 m (3) 1 km 386 m
(4) 6 km 34 m (5) 8 km 324 m (6) 7 km 945 m
B. (1) 4 m (2) 7 m 50 cm (3) 3 m 85 cm (4) 8 m 10 cm
(5) 2 m 5 cm (6) 5m 67 cm (7) 6 m 84 cm (8) 9 m 98 cm
C. (1) 3 cm 5 mm (2) 63 cm 4 mm (3) 59 cm 3 mm (4) 40 cm
(5) 29 cm 5mm (6) 44 cm 7 mm (7) 60 cm 9 mm (8) 89 cm 9 mm
D. (1) 8000 m (2) 20340 m (3) 1500 cm (4) 2545 cm
(5) 350 mm (6) 1 km 200 m (7) 3 km 785 m (8) 15 m 20 cm
(9) 85 cm (10) 472 cm 5 mm
E. (1) 7m (2) 8 m (3) 9 m (4) 40 m
(5) 10 m (6) 50m (7) 6 m (8) 100 m

EXERCISE 5.2

- A. (1) 109 km 72 m (2) 685 m 44 cm (3) 8 km 80 m
(4) 14 m 30 cm (5) 5 km 242 m (6) 111 cm 5 mm
(7) 16 km 180 m (8) 14 m 22 cm

ANSWERS

- B.** (1) 8 km 991 m (2) 56 m 3 cm (3) 1 km 742 m
(4) 23 cm 5 mm (5) 2 km 420 m (6) 22 m 37 cm
(7) 2 km 805 m (8) 5 cm 3 mm

EXERCISE 5.3

- (1) 3 m 92 cm ribbon left (2) 1 km 457 m total distance
(3) 79 cm space left (4) Railway station, 129 m
(5) 12 cm 2 mm length (6) 117 m ahead
(7) 116 km drive (8) 1 m 5 cm tall

EXERCISE 5.4

- A.** (1) 120 minutes (2) 480 minutes (3) 720 minutes
(4) 1620 minutes (5) 1800 minutes (6) 2040 minutes
- B.** (1) 300 seconds (2) 600 seconds (3) 1200 seconds
(4) 2700 seconds (5) 3300 seconds (6) 3600 seconds
(7) 4500 seconds (8) 5100 seconds
- C.** (1) 19 hours 40 minutes (2) 20 hours 50 minutes (3) 24 hours 50 minutes
(4) 37 hours 5 minutes (5) 30 hours 15 minutes (6) 39 hours 35 minutes
- D.** (1) 36 minutes 25 seconds (2) 4 minutes 35 seconds
(3) 5 minutes 50 seconds (4) 11 minutes 50 seconds
(5) 16 minutes 30 seconds (6) 23 minutes 15 seconds
- E.** (1) 1 hour 3 minutes 20 seconds (2) 1 hour 8 minutes 20 seconds
(3) 1 hour 12 minutes 40 seconds (4) 1 hour 16 minutes 35 seconds
(5) 1 hour 18 minutes 45 seconds (6) 1 hour 21 minutes 55 seconds

EXERCISE 5.5

- A.** (1) 51 minutes 20 seconds (2) 67 minutes 14 seconds
(3) 8 hours 37 minutes (4) 6 hours 16 minutes 16 seconds
(5) 6 hours 44 minutes 29 seconds
- B.** (1) 23 minutes 41 seconds (2) 35 seconds
(3) 7 minutes (4) 2 hours 28 minutes
(5) 1 hour 44 minutes 50 seconds

ANSWERS

EXERCISE 5.6

- | | |
|-----------------------|------------------------|
| (1) 6 weeks 4 days | (2) 11 weeks 3 days |
| (3) 30 weeks 3 days | (4) 334 weeks 5 days |
| (5) 15 months | (6) 26 months 20 days |
| (7) 23 months 20 days | (8) 21 months 20 days |
| (9) 5 years 6 months | (10) 6 years 10 months |
| (11) 4 years 1 month | (12) 20 years 4 months |

EXERCISE 5.7

- | | |
|----------------------------------|------------------------|
| (1) Ali plays 20 minutes shorter | (2) 2 hours 10 minutes |
| (3) 3 months 20 days | (4) 2 hours 26 minutes |
| (5) 1 hour 54 minutes | |

EXERCISE 5.8

- | | | | | | |
|----|------------|------------|-----------|-----------|------------|
| A. | (1) 20°C | (2) 35°C | (3) 40°F | (4) 85°F | |
| B. | (1) 5°C | (2) 25°C | (3) 35°C | (4) 20°C | (5) 95°C |
| | (6) 110°C | (7) 125°C | (8) 135°C | (9) 50°C | (10) 75°C |
| | (11) 110°C | (12) 120°C | | | |
| C. | (1) 86°F | (2) 113°F | (3) 185°F | (4) 131°F | (5) 194°F |
| | (6) 50°F | (7) 68°F | (8) 140°F | (9) 176°F | (10) 230°F |

EXERCISE 5.9

- | | | |
|--------------------|--------------------------|----------|
| (1) 95°F | (2) Jacobabad 104°F more | (3) 40°C |
| (4) 9°F difference | (5) 18°F difference | |

REVIEW EXERCISE 5

- | | | | | |
|----|---------------------------------|--------------------------------|--------------|--------------|
| 1. | (i) 28648 m | (ii) 4897 cm | (iii) 769 mm | (iv) 6758 mm |
| 2. | (i) 1 hour 4 minutes 37 seconds | (ii) 1 day 13 hours 13 minutes | | |
| | (iii) 8 m 51 cm 4 mm | | | |
| 3. | (i) 44 km 577 m | (ii) 1 m 27 cm 4 mm | | |
| | (iii) 36 hours 43 minutes | | | |
| 4. | (i) b | (ii) b | (iii) d | (iv) a |
| 5. | (i) 12:35 | (ii) 10:30 | | |
| 6. | (i) 45 minutes | (ii) 1 hour 5 minutes | | |

ANSWERS

EXERCISE 6.1

- | | | | |
|--------------|---------------|-------------|--------------|
| (1) Rs 180 | (2) Rs 528.50 | (3) Rs 770 | (4) Rs 60000 |
| (5) Rs 5.5 | (6) Rs 40 | (7) Rs 4.50 | (8) Rs 20.50 |
| (9) Rs 12800 | (10) Rs 62.80 | | |

EXERCISE 6.2

- | | | | |
|---------------|---------------|---------------|--------------------------|
| (1) Rs 400 | (2) Rs 360 | (3) Rs 105 | (4) $7\frac{1}{2}$ hours |
| (5) Rs 48000 | (6) 4680 km | (7) 14 shirts | (8) 936 passengers |
| (9) 1163.4 kg | (10) Rs 29880 | | |

EXERCISE 6.3

- | | | | |
|--------------------------|-----------------|---------------|-------------|
| 1. (i) F | (ii) T | | |
| 2. (i) 1:2 | (ii) 16:5 | (iii) 5:21 | (iv) 500:3 |
| 3. (i) Direct | (ii) Inverse | (iii) Inverse | (iv) Direct |
| 4. Rs 48 | 5. 26.25 km | 6. 18 workers | |
| 7. $7\frac{1}{2}$ litres | 8. 120 soldiers | 9. 66 farmers | |

REVIEW EXERCISE 6

- | | | | |
|-----------|----------------|------------|-----------|
| 1. Rs 54 | 2. Rs 650 | 3. Rs 90 | 4. 4 days |
| 5. Rs 120 | 6. 5000 papers | 7. 2000 km | |

EXERCISE 7.1

- | | | |
|--------------------|--------------------|-----------------|
| 2. (a) acute angle | (b) obtuse angle | (c) right angle |
| (d) reflex angle | (e) straight angle | (f) acute angle |
| (g) reflex angle | (h) obtuse angle | (i) right angle |

EXERCISE 7.2

- | | | | |
|--------------------|------------------|-------------------|------------------|
| 1. (i) 335° | (ii) 330° | (iii) 270° | (iv) 280° |
|--------------------|------------------|-------------------|------------------|

ANSWERS

EXERCISE 7.3

1. (i) Yes (ii) Yes (iii) No (iv) No
2. No, because they do not have common vertex
3. (i) 30° (ii) 14° (iii) 45° (iv) 52° (v) 75°
4. (i) 155° (ii) 135° (iii) 110° (iv) 82° (v) 37°
5. (i) Complementary (ii) Supplementary (iii) Supplementary
(iv) Complementary (v) Supplementary (vi) Complementary
6. (i) 45° (ii) 90°
7. (i) No (ii) No (iii) Yes

EXERCISE 7.7

1. (i) Rhombus (ii) Kite (iii) Trapezium (iv) Kite

REVIEW EXERCISE 7

1. Right angle
2. (i) Straight angle (ii) Right angle (iii) Straight angle
4. (i) No (ii) Yes (iii) No (iv) Yes
5. Three; AEDF, BEFD, DEFC
6. Equilateral triangles ABC, FGE, GEC, ECD, AFE, BFG
Trapeziums FEBC, FDGC, FACG, BAGE
Parallelogram FDCB
Rhombus EDGC, FEBG, FAEG
7. 70°

EXERCISE 8.1

- A. (i) ✗ (ii) ✓ (iii) ✓ (iv) ✓ (v) ✗
(vi) ✓ (vii) ✓ (viii) ✓ (ix) ✗
- B. (1) (a) cm (b) sq. cm
(2) (a) cm (b) sq. cm
(3) (a) m (b) sq. m
- C. (1) (a) cm (b) sq. cm (2) (a) cm (b) sq. cm
(3) (a) cm (b) sq. cm (4) (a) cm (b) sq. cm

ANSWERS

EXERCISE 8.2

- A.** (1) 7 cm (2) 100 mm (3) 80 mm
- B.** (1) 640000 sq. cm (2) 980100 sq. cm
(3) 592900 sq. cm (4) 1518000 sq. cm
- C.** (1) area = 6 sq. cm, perimeter = 10 cm
(2) area = 5 sq. cm, perimeter = 12 cm
(3) area = 12 sq. cm, perimeter = 14 cm
(4) area = 16 sq. cm, perimeter = 20 cm
(5) area = 45 sq. cm, perimeter = 28 cm
(6) area = 28 sq. cm, perimeter = 22 cm
(7) area = 9 sq. cm, perimeter = 13 cm
(8) area = 28 sq. cm, perimeter = 23 cm
- D.** (1) area = 16 sq. cm, perimeter = 16 cm
(2) area = 36 sq. cm, perimeter = 24 cm
(3) area = 56.25 sq. cm, perimeter = 30 cm
(4) area = 67.24 sq. cm, perimeter = 32.8 cm
(5) area = 3136 sq. mm, perimeter = 224 mm
(6) area = 8464 sq. mm, perimeter = 368 cm
- E.** (1) E (2) F (3) A, B, C, D and F
(4) E (5) A and D, B and E, C and F

EXERCISE 8.3

- (1) 280 m perimeter (2) 49 sq. m area
(3) 240 cm long (4) Area = 5400 sq. m, perimeter = 300 m
(5) 860 cm lace (6) Area of square = 625 sq. m
(7) Area of rectangle = 600 sq. m
(8) (i) 180 sq. m (ii) 100 sq. m (iii) Floor, 80 sq. m

REVIEW EXERCISE 8

- A.** (1) d (2) c (3) c
(4) c (5) c (6) b
- B.** (1) 4 x side (2) 2 x (length + breadth)
(3) perimeter = 28 cm, area = 49 sq. cm
(4) perimeter = 26 cm, area = 40 sq. cm

ANSWERS

EXERCISE 9.1

- (1) 16 (2) 4 (3) 7.875 (4) 9.625
(5) $\frac{49}{80}$ (6) $\frac{1009}{3000}$ (7) 3.85 (8) 8.33
(9) $7\frac{763}{900}$ (10) $6\frac{11}{12}$

EXERCISE 9.2

- (1) Average marks = 68.4 (2) Average Expenditure per day = Rs 19
(3) Average temperature = 40.6°C (4) 2 paras
(5) Average score = 70 (6) Average income = Rs 634
(7) Average speed = 70 km (8) Average speed = 47.5 km
(9) Average runs = 9 (10) Average rain fall = 8 mm

EXERCISE 9.3

2. (i) 15 millions (ii) 30 millions (iii) 35 millions
(iv) 1990 (v) 2010 (vi) 5 millions
3. (i) February (ii) May (iii) Rs. 5500
(iv) Rs. 7000 (v) January and April (vi) March
(vii) Rs. 10000 (viii) Rs. 3000

REVIEW EXERCISE 9

1. (i) 6 (ii) 30
2. 19 km
3. 80 runs
4. (i) 3 (ii) 5 (iii) 7 (iv) 2 (v) 2 (vi) 9
5. (i) 30 (ii) 40 (iii) 20 (iv) 60 (v) 70 (vi) 80
6. (i) 16 (ii) 31.25 (iii) Bar